SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-033: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR-033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	V. Susmitha
Student ID	JFICT24_30
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators can currently view student account and profile information through the Student Management System. However, the administrator Students table does not display the date and time of a student's most recent successful login.

CR-033 was introduced to improve account activity monitoring by recording the last successful login timestamp and displaying it in the administrator Students table. This enables administrators to identify recently active student accounts and students who have never logged in.

1.2 Existing Behaviour

Before implementation:

- The login process validates the username and password and generates a JWT token after successful authentication.
- The users table stores account and role information but does not contain a last login timestamp.
- The administrator Students table displays student profile information, including Registration No., Name, Email, Phone, and Actions.
- Administrators cannot identify when a student last successfully logged in.
- Students who have never logged in are not explicitly identified in the Students table.

1.3 Requested Change

The requested change was to introduce last login tracking and display functionality for student accounts.

The implementation requirements were:

- Add a nullable `last_login` DATETIME column to the users table.
- Update the timestamp only after successful username and password verification.
- Use the MySQL `NOW()` function to record the current database date and time.
- Update the existing student model queries to retrieve the last login timestamp.
- Extend the existing `GET /api/students/` response to include `last_login`.
- Add a Last Login column to the administrator Students table.
- Format the timestamp into a readable date and time.
- Display "Never logged in" when the timestamp is NULL.
- Preserve existing student Add, Edit, Delete, authentication, and authorization functionality.

2. Implementation

2.1 Implementation Summary

CR-033 was implemented across the database, backend authentication flow, student data retrieval, and administrator frontend.

A nullable `last_login` column was added to the users table to store the most recent successful login date and time.

A new `updateLastLogin()` function was introduced in `backend/models/userModel.js`. The function updates the timestamp using the MySQL `NOW()` function. In `backend/controllers/authController.js`, the timestamp is updated only after the submitted password has been successfully verified using bcrypt.

The existing student retrieval functionality was updated in `backend/models/studentModel.js` to join the students and users tables and retrieve the corresponding `last_login` value.

Finally, `frontend/src/pages/DashboardPage.jsx` was updated to display a Last Login column in the administrator Students table. The frontend formats available timestamps into a readable date and time and displays "Never logged in" when no successful login has been recorded.

The existing login response, JWT generation, student management operations, and authorization mechanisms were preserved.

2.2 Affected Files

File	Change	Purpose
`database/schema.sql`	Modified	Add the nullable `last_login` column to the users table.
`backend/models/userModel.js`	Modified	Add the `updateLastLogin()` function to update the timestamp.
`backend/controllers/authController.js`	Modified	Update the last login timestamp after successful password verification.
`backend/models/studentModel.js`	Modified	Retrieve `users.last_login` with student records using a JOIN.
`frontend/src/pages/DashboardPage.jsx`	Modified	Add the Last Login column and format the timestamp for display.

2.3 Technical Changes

Database Modification

The users table was updated to include a nullable `last_login` DATETIME column.

The column stores the date and time of the most recent successful login. The NULL value indicates that no successful login has been recorded for the account.

The database timestamp is updated using the MySQL `NOW()` function.

Backend Authentication Update

The existing login controller was modified to update the last login timestamp after successful password verification.

The authentication process follows these steps:

1. Validate the submitted username and password.
2. Retrieve the user account using the existing user model.
3. Verify the submitted password using bcrypt.
4. Reject the login if the credentials are invalid.
5. Update the user's `last_login` timestamp after successful password verification.
6. Continue the existing JWT generation and login response process.

This ensures that unsuccessful login attempts do not update the last login timestamp.

Student API Update

The existing student retrieval functionality was updated to retrieve the `last_login` field from the related users record.

The student model uses a JOIN between the students and users tables based on `user_id`.

The existing `GET /api/students/` endpoint continues to provide the student list, with the response extended to include the `last_login` field. No new API endpoint was required.

Frontend Modification

The administrator dashboard was updated to display the last login information.

The following frontend changes were implemented:

- Added a Last Login column to the Students table.
- Retrieved the timestamp from each student's API response.
- Formatted the timestamp into a readable date and time.
- Displayed “Never logged in” when the value is NULL or unavailable.
- Preserved the existing Profile, Registration No., Name, Email, Phone, and Actions columns.
- Preserved the original `student.student_id` for Edit and Delete operations.

Security and Data Handling

The implementation preserves the existing authentication and authorization mechanisms.

- Password verification continues to use bcrypt.
- Failed login attempts do not update `last_login`.
- Existing JWT generation and expiration behaviour remain unchanged.
- Password hashes and other sensitive authentication information are not added to the student API response.
- Existing administrator authorization rules remain unchanged.

2.4 Implementation Logic

The Last Login feature operates as follows:

1. The administrator opens the Students page.
2. The frontend requests student records through the existing `GET /api/students/` endpoint.
3. The backend retrieves student information together with the associated `last_login` value.
4. The frontend displays the returned records in the Students table.
5. If a student has a recorded login timestamp, the date and time are formatted for readable display.
6. If the timestamp is NULL or unavailable, the frontend displays “Never logged in”.
7. When a student successfully logs in, the backend updates the corresponding user's `last_login` timestamp.
8. Subsequent student list requests retrieve the latest recorded timestamp.

2.5 Error and Data Handling

The existing login validation and error handling were preserved.

Invalid credentials continue to produce the existing authentication error response. The last login timestamp is not updated when the password verification fails.

The nullable database field allows student accounts without a recorded successful login to be handled correctly.

The frontend handles missing login timestamps by displaying “Never logged in” instead of an empty value.

Existing student loading, authentication error handling, and CRUD operations remain unchanged.

3. Testing and Verification

3.1 Testing Approach

The implementation was evaluated against the functional, security, and regression requirements of CR-033.

Testing Types:

- Functional testing
- Positive testing
- Negative testing
- Database testing
- API testing
- Authentication testing
- UI testing
- Regression testing
- CRUD verification

Environment: Development

Test Role: Administrator and Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	Successful student login	`last_login` is updated with the current database date and time.	Timestamp updated as expected.	Pass
TC02	Login with an incorrect password	Login is rejected and `last_login` remains unchanged.	Login rejected and timestamp unchanged.	Pass
TC03	Load the administrator Students table	The student API returns the `last_login` field with student records.	Last Login data retrieved.	Pass
TC04	Display a recorded login timestamp	The timestamp is displayed in a readable date and time format.	Formatted timestamp displayed.	Pass
TC05	Display a student account without a recorded login	The table displays “Never logged in”.	Fallback message displayed.	Pass
TC06	Perform multiple successful logins	The timestamp reflects the most recent successful login.	Latest timestamp displayed.	Pass
TC07	Edit a student record	The correct student can be edited using `student_id`.	Existing Edit behaviour preserved.	Pass
TC08	Delete a student record	The correct student can be deleted without breaking the student list.	Existing Delete behaviour preserved.	Pass
TC09	Add a student record	The existing Add Student functionality continues to work.	Existing Add behaviour preserved.	Pass
TC10	Verify existing authentication and authorization	Existing login, JWT authentication, and administrator access controls continue to work.	Existing behaviour preserved.	Pass

Note: The test results above should be retained as Pass only for test cases actually executed and verified in the development environment.

3.3 Test Summary

Test Result	Count
Total Test Cases	10
Passed	10
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

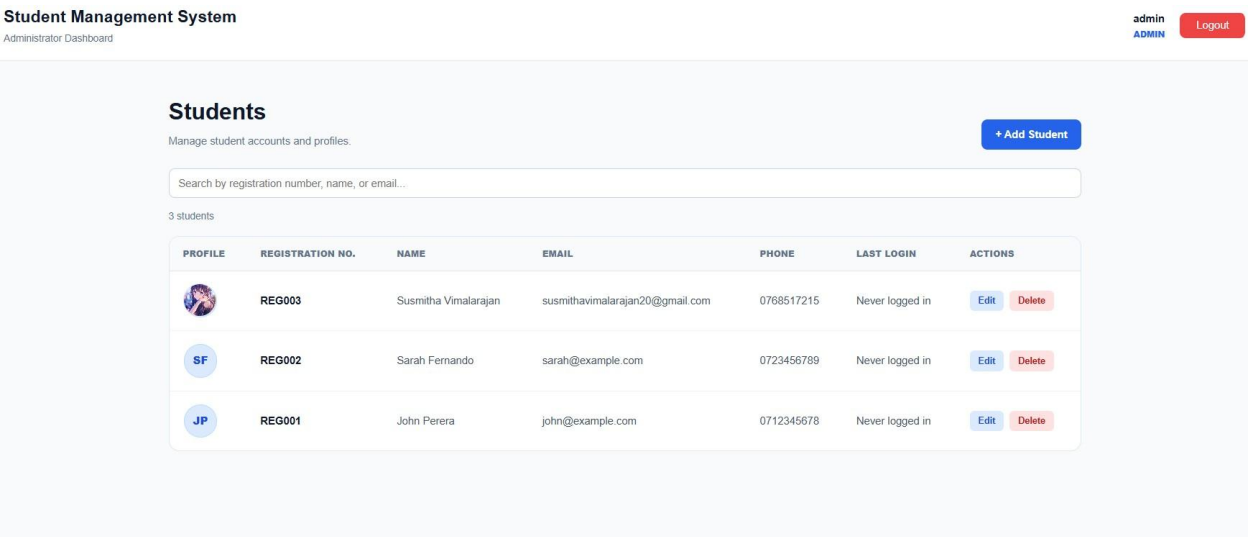
ID	Acceptance Criterion	Verification Method	Status
AC-001	A successful login records the student's latest login date and time.	Successful login test	Verified
AC-002	Incorrect login credentials do not update the last login timestamp.	Invalid password test	Verified
AC-003	The existing student API returns the `last_login` field.	API response verification	Verified
AC-004	The administrator Students table displays a Last Login column.	Frontend UI verification	Verified
AC-005	Students without a recorded login display “Never logged in”.	NULL timestamp test	Verified
AC-006	The displayed timestamp is formatted into a readable date and time.	Frontend display verification	Verified
AC-007	Existing Add, Edit, and Delete operations continue to work correctly.	CRUD regression testing	Verified
AC-008	Existing authentication and authorization behaviour remains unchanged.	Authentication and authorization testing	Verified
AC-009	Password hashes and other sensitive authentication information are not exposed through the student API.	API response inspection	Verified

5. Evidence

A maximum of two relevant screenshots should be attached.

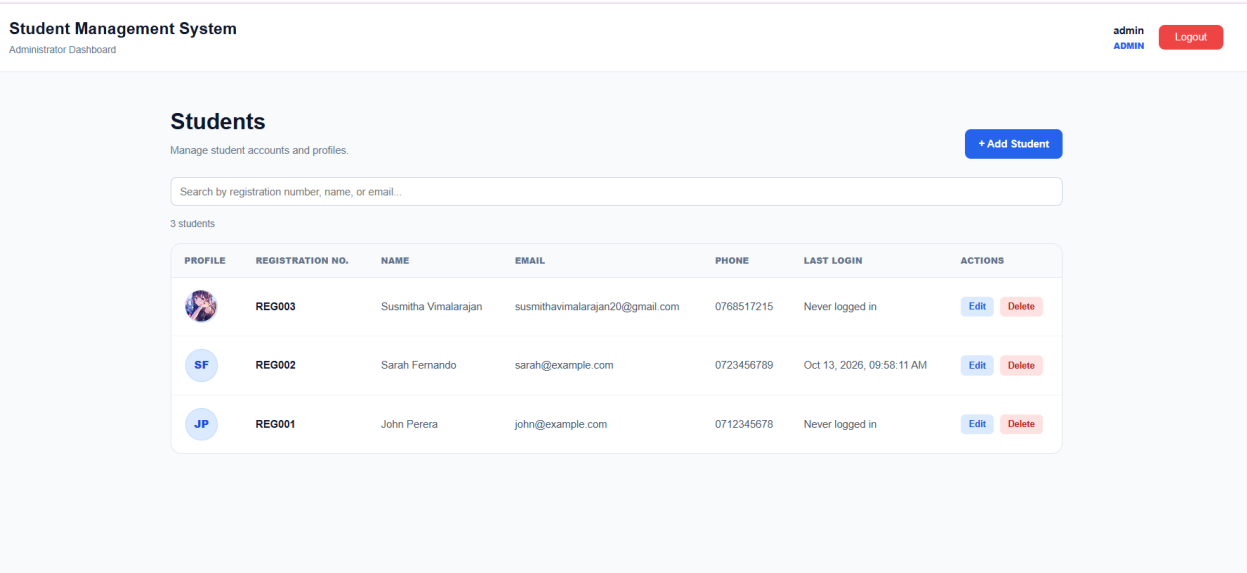
Evidence 01 – Last Login Column

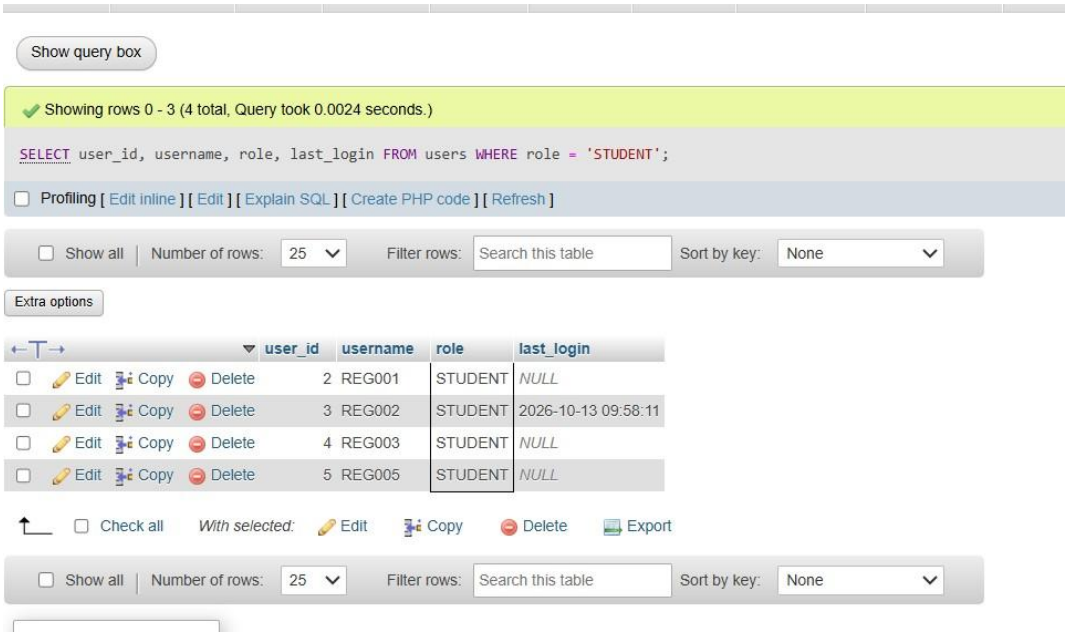
Related Acceptance Criteria: AC-003, AC-004, AC-005, AC-006



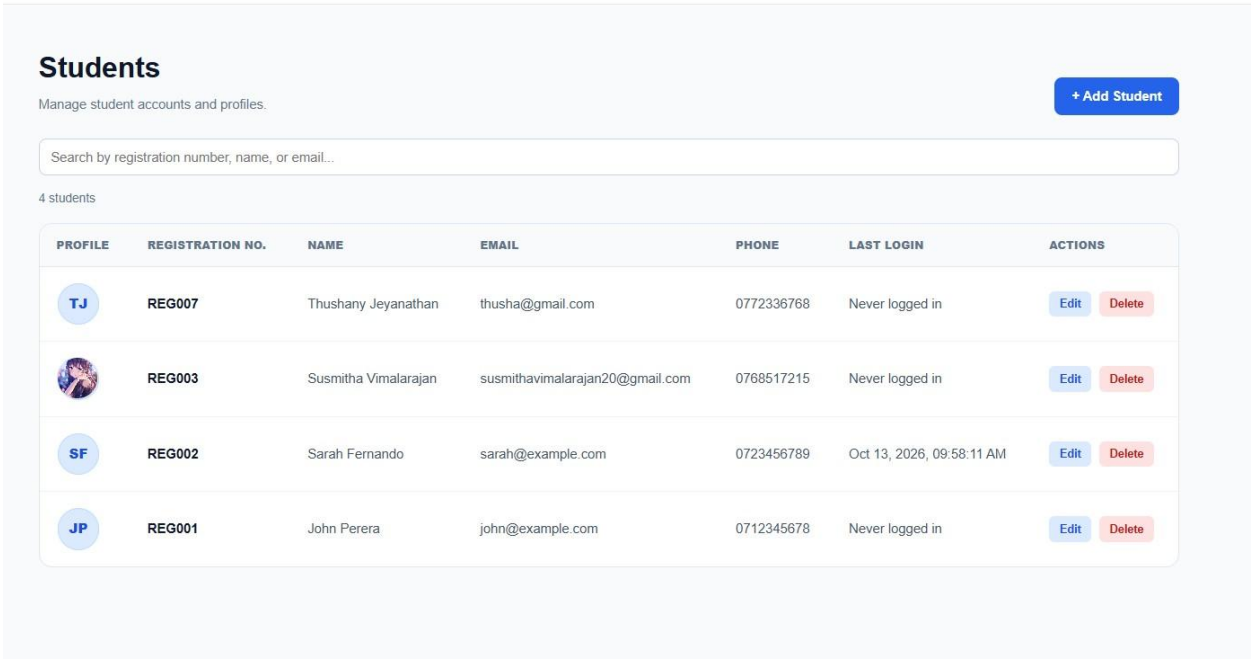
Evidence 02 – Login Tracking and CRUD Verification

Related Acceptance Criteria: AC-001, AC-002, AC-007, AC-008





After create new student,



Students

Manage student accounts and profiles.

+ Add Student

Search by registration number, name, or email...

4 students

PROFILE	REGISTRATION NO.	NAME	EMAIL	PHONE	LAST LOGIN	ACTIONS
TJ	REG007	Thushany Jeyanathan	thusha@gmail.com	0772336768	Oct 13, 2026, 10:47:24 AM	Edit Delete
	REG003	Susmitha Vimalarajan	susmithavimalarajan20@gmail.com	0768517215	Never logged in	Edit Delete
SF	REG002	Sarah Fernando	sarah@example.com	0723456789	Oct 13, 2026, 09:58:11 AM	Edit Delete
JP	REG001	John Perera	john@example.com	0712345678	Never logged in	Edit Delete

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing CRUD	Verified
Final Status	Completed

6.1 Final Summary

CR-033 was implemented to introduce Last Login tracking and display functionality for student accounts in the Student Management System.

The implementation adds a nullable `last_login` column to the users table and updates the timestamp after successful password verification. The existing student model retrieves the timestamp through a JOIN between the students and users tables, and the existing student API returns the additional field without requiring a new endpoint.

The administrator dashboard displays the Last Login column with a readable date and time. Accounts without a recorded successful login display “Never logged in”.

The implementation preserves the existing bcrypt password verification, JWT authentication flow, authorization rules, and student Add, Edit, and Delete functionality.

The change improves administrator visibility of student account activity while maintaining the existing system functionality.

6.2 Issues and Limitations

No significant issues have been identified for the requested Last Login feature, subject to final regression testing.

The feature records only the most recent successful login timestamp. It does not maintain a history of previous login attempts or login sessions, as these functions are outside the scope of CR-033.

7. Student Declaration

Student Name: V. Susmitha

Student ID: JFICT24_30

Date: 13 October 2026

Signature: Susmitha